06-04 第1篇

阅读理解（说明文） Why Do Our Fingers Get Wrinkled in Water?

① After a long bath or forty minutes of washing dishes, if you look closely at the tips of your fingers, you will find that they have turned into small, soft mountain ranges. The skin is covered with tiny lines and folds. Most people have noticed this change many times, but few have asked a simple question: **why does our skin, strong as it normally is, crumple in water?** For many years scientists believed the answer was simple. Today we know it is more interesting.

② The old idea was that our skin "absorbs" water like a sponge. When the outside layer soaks up water, it swells. Because this layer is glued to the layer below, it cannot stretch outward; instead, it folds, creating wrinkles. This sounds reasonable, but it does not fully explain one surprising fact: **people whose nerves in the hand are damaged do not get wrinkled fingers, even after a long bath**. If wrinkles were only the result of water soaking, damaged nerves would not matter.

③ In 1936, a doctor noticed this strange pattern: patients with nerve injuries had smooth fingers in water, while their healthy hand still wrinkled as usual. Since then, many researchers have tested the idea. Today most scientists accept a newer explanation: fingertip wrinkling is partly controlled by the **nervous system**. When the hand is in water for several minutes, tiny nerves send a signal to the small blood vessels under the skin, telling them to shrink. As blood vessels become narrower, the skin above is pulled into folds — the wrinkles we see.

④ If wrinkles are not simply caused by water, why has our body developed this automatic response? One popular answer comes from a **hand-grip** experiment. Researchers asked volunteers to pick up wet marbles with both wrinkled and smooth fingers. The result was clear: wrinkled fingers gripped the wet marbles faster and more firmly. The tiny lines, like the patterns on car tyres, let water run away more easily, which increases friction and grip. In short: wrinkles may help us hold on to wet things.

⑤ This idea fits our history. Long ago, our ancestors needed to collect food from wet riverbanks or from fishing nets. Hands that held slippery things better would help a person survive. Over many generations, bodies with this "rain-tyre" ability may have been kept by nature. Each wrinkle, then, is not a sign that our skin is weak, but a gentle reminder that our hands are still very, very old tools.

⑥ Next time your fingers turn into soft little maps after a long swim, smile. Inside those strange folds, you can read a short story about nerves, blood vessels and wet ancient

riverbanks — a story kept by your own body, quietly, across thousands of years.

根据短文内容，选择最佳答案。

1. What is the passage mainly about?

A. Why we should wash our hands.
B. Why our fingers get wrinkled in water.
C. How to cure nerve damage.
D. How to wash clothes faster.

2. What was the old idea about why fingers wrinkle in water?

A. The skin becomes angry in water.
B. The skin absorbs water like a sponge, swells, and then folds.
C. The water has special colour.
D. The heart beats faster.

3. What strange fact does the old idea fail to explain?

A. Cats have wrinkled fingers.
B. People whose hand nerves are damaged do not get wrinkled fingers, even after a long bath.
C. Fingers never wrinkle.
D. Wrinkles turn red in cold water.

4. According to the newer explanation, how do wrinkles really form?

A. Nerves send a signal to tiny blood vessels under the skin; when the vessels shrink, the skin is pulled into folds.
B. The water softens the bone.
C. The skin is pushed by the heart directly.
D. Small balloons under the skin explode.

5. What did the "hand-grip" experiment suggest about wrinkled fingers?

A. They make our hands slower.
B. They help people hold wet things more firmly, like the pattern on car tyres helps on a wet road.
C. They make people feel cold more quickly.
D. They cause skin diseases.

答案与解析

1.
【答案】B
【题型】主旨大意题
【定位】全文
【解析】文章解释手指泡水后起皱的原因。A/C/D 均非主题。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第②段 "skin 'absorbs' water like a sponge..."
【解析】旧说法：吸水像海绵。A/C/D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第②段末及第③段
【解析】神经受损者手指不起皱。A/C/D 均错。故选 B。

4.
【答案】A
【题型】细节理解题
【定位】第③段
【解析】神经控制血管收缩引起皱纹。B/C/D 均错。故选 A。

5.
【答案】B
【题型】细节理解题
【定位】第④段 "wrinkled fingers gripped the wet marbles faster and more firmly... like the patterns on car tyres..."
【解析】皱纹提高抓湿物的能力。A/C/D 均错。故选 B。

词汇注释（超纲词）

- **wrinkle** [n./v.] 皱纹 / 起皱
- **nerve** [n.] 神经
- **blood vessel** [n.] 血管
- **friction** [n.] 摩擦
- **ancestor** [n.] 祖先